

# Knowing You Don't Know:

## Learning When to Continue Search in Multi-round RAG through Self-Practicing

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Code:



Paper:



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# Motivation: Why RAG Gets It Wrong

Q

What is the 4th highest peak in North America?



0 Round RAG (no retrieval)

**ANSWER**

Mount Foraker

over-confident (x)



1 Round RAG

**Round 1 Docs:** Denali is the highest mountain peak in North America, with a summit elevation of 20,310 feet (6,190 m) above sea level...

**ANSWER**

Mount Bona

over-confident (x)



2 Rounds RAG

**Round 1 Docs:** Denali...

**Round 2 Docs:** Mount Saint Elias: Rising to an elevation of 18,008 feet (5,489 m) above Icy Bay, it is the second tallest peak in both the United States and Canada and fourth tallest in North America ...

**ANSWER**

Mount Saint Elias



3 Rounds RAG

**Round 1 Docs:** Denali...

**Round 2 Docs:** Mount Saint Elias ...

**Round 3 Docs:** Mount Bona: As the 10 tallest peaks in the US are in Alaska, Mount Bona is naturally the 4th highest peak in both. The summit has an ...

**ANSWER**

Mount Bona

over-retrieval (x)

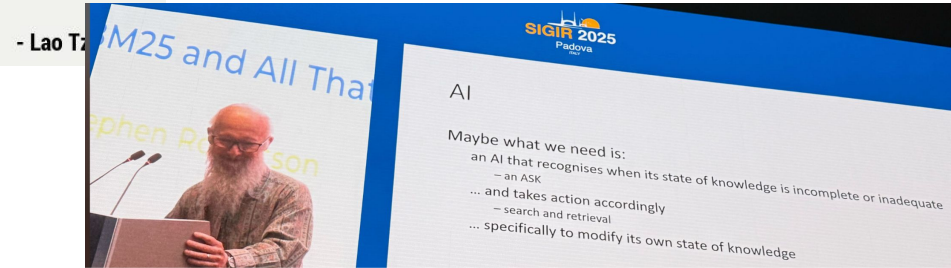
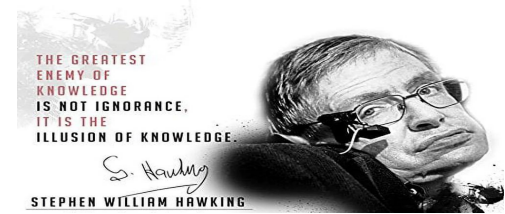
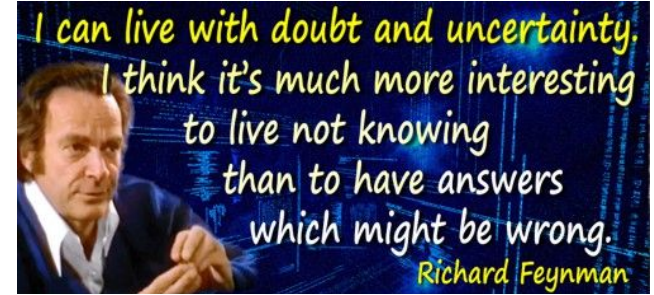
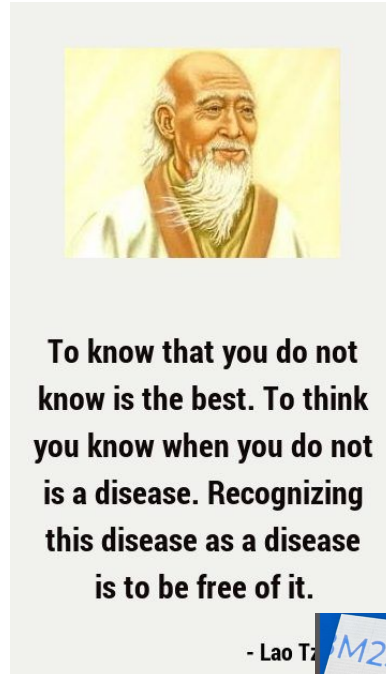
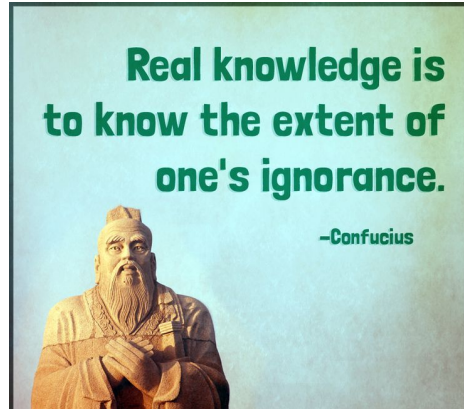
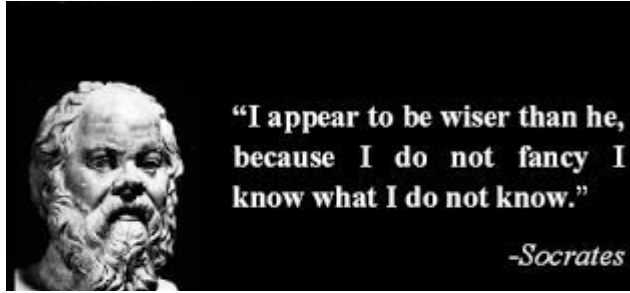
## Over-Confident

answering the question  
without sufficient information

## Poor Retrieval Control

stop earlier or retrieve too much  
unnecessary information which may  
confuse the LLM

# Knowing You Don't Know: L'uomo più saggio è colui che sa di non sapere

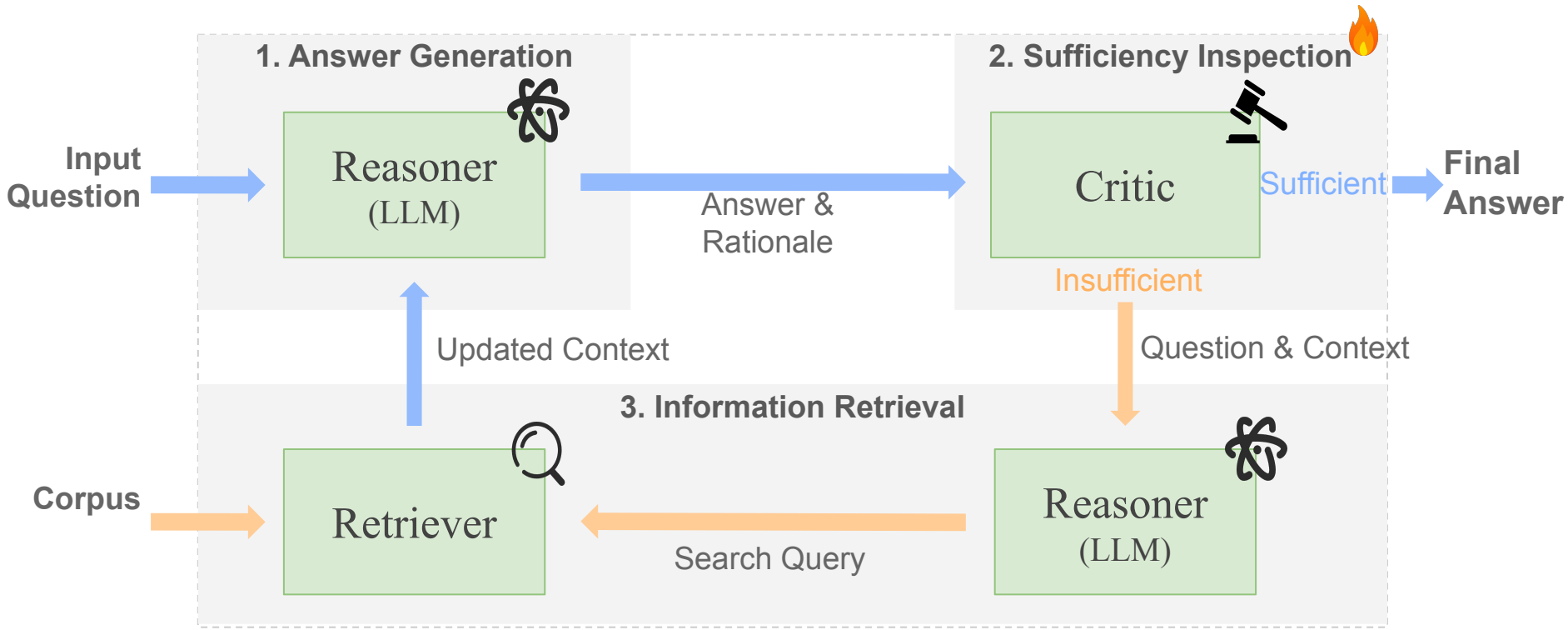


# **Proposed Solution: Self-Practicing for Inner Monologue-based RAG (SIM-RAG)**

Main idea: Train a Critic Model to Judge RAG/LLM's Output and Reasoning Process (Inner Monologue)

- Step 1: Self-Practice — Generating Training Data
- Step 2: Critic Model Learning from Reasoning Trials
- Step 3: Inference Time: Multi Round SIM-RAG with Critic in Action

**Step 3.** Inference Time: Multi Round SIM-RAG with Critic in Action

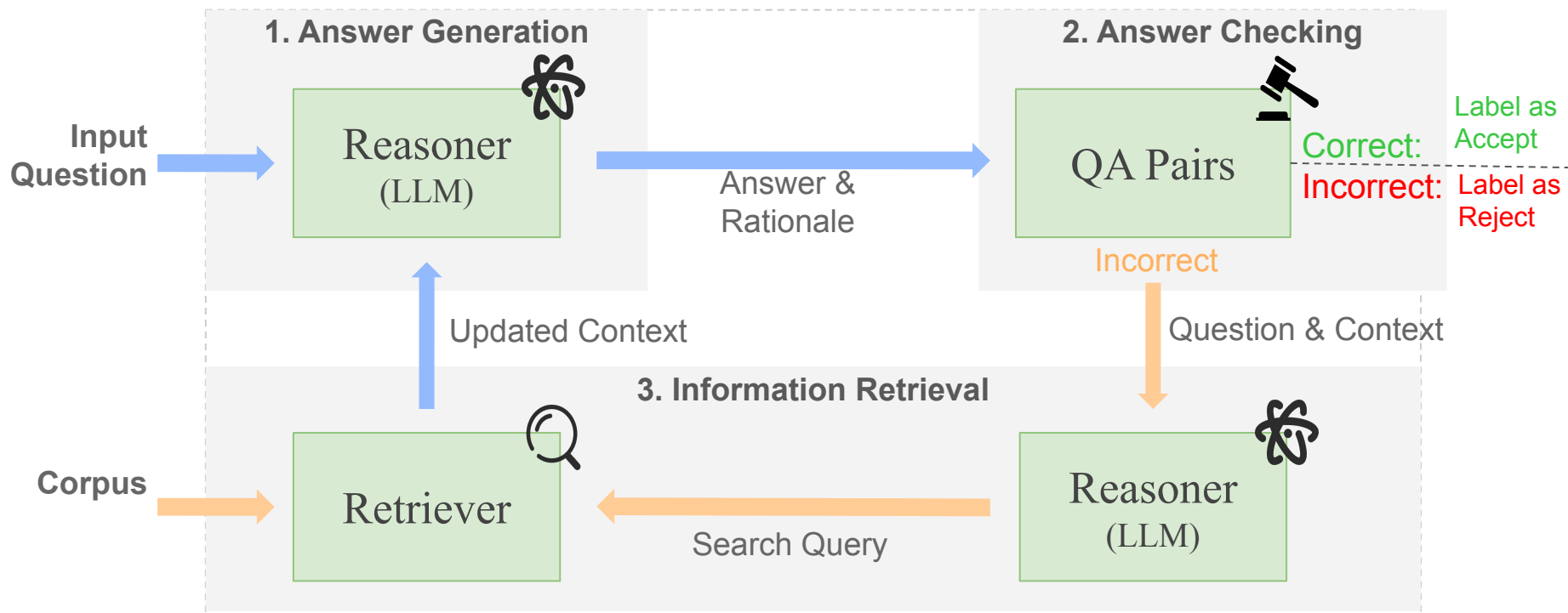


# Step 1. Self-Practice — Generating Training Data

## Inner Monologue Distillation and Labeling

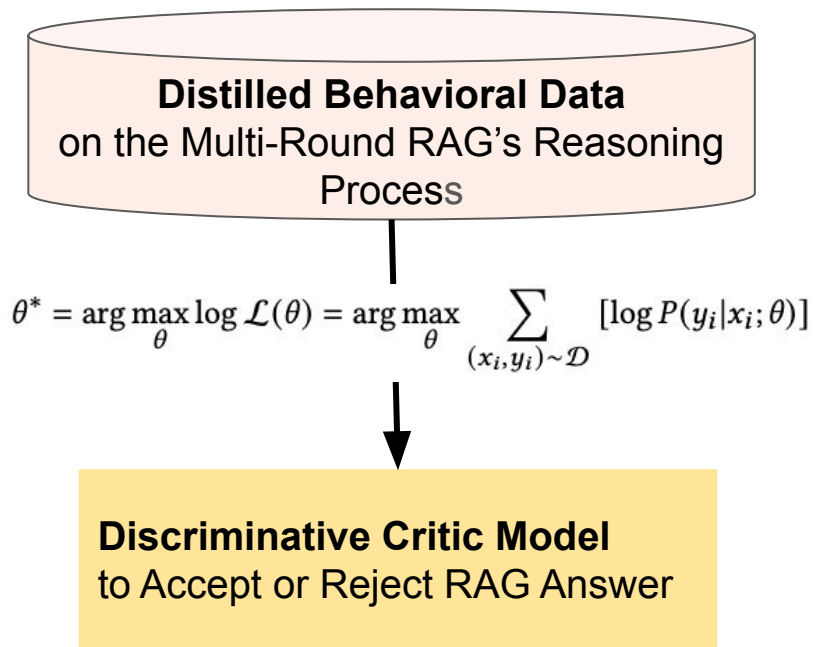
Given: Question Answer Pairs as Seed

Results: High-Quality Positive and Negative Procedure Training Data



## Step 2. Critic Model Learning from Reasoning Trials

**Learn RAG behavior** by training a lightweight classifier



# SIM-RAG Design Decisions

- **Easy Integration with Any Existing RAG Pipeline:** No need for RAG-LLM fine-tuning — works with open- and closed-source models without hurting alignment.
- **Avoid Self-Bias and Overconfidence of LLM:** A separate Critic as an independent reviewer
- **Low Cost Training with Self-Practice:** Cost-effective data, lightweight model tuning

Does **Knowing You Don't Know...** Work?

Experimental Results



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# Best SIM-RAG Outperforms All Baselines

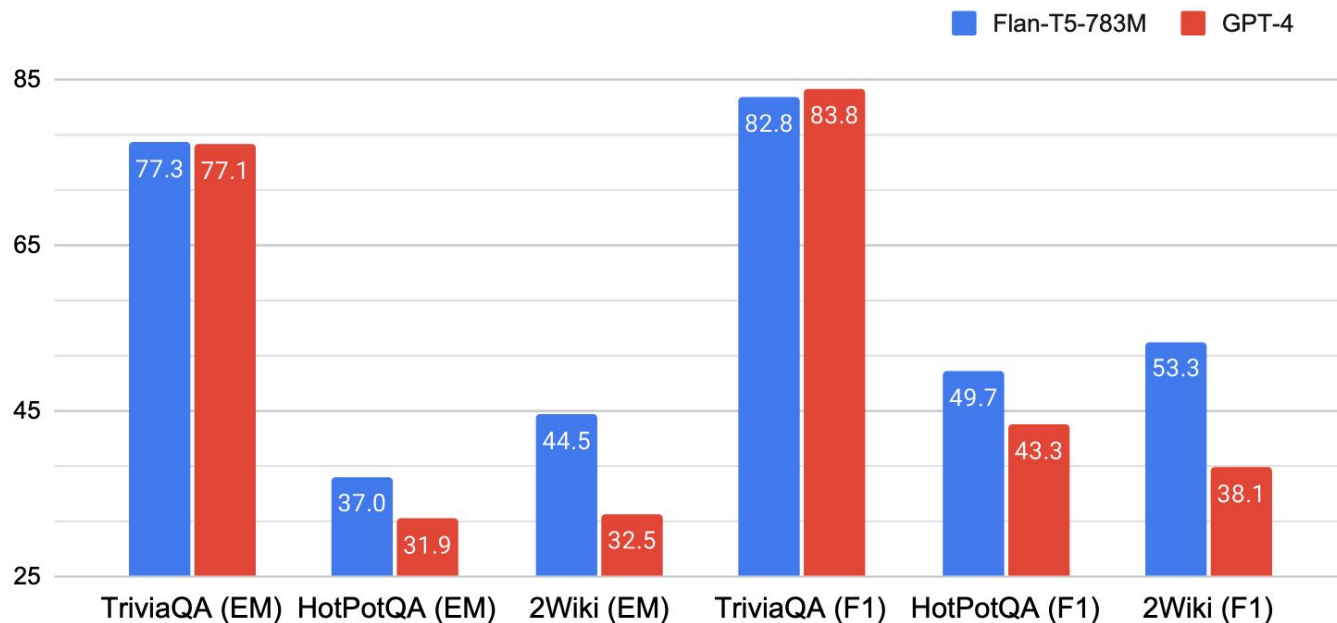
|                |                  | TriviaQA     |      | HotPotQA |      | 2Wiki |      |      |
|----------------|------------------|--------------|------|----------|------|-------|------|------|
| Method         | LLM/Retriever    | EM           | F1   | EM       | F1   | EM    | F1   |      |
| (a)            | Prompting        |              |      |          |      |       |      |      |
|                | NaiveGen [14]    | Llama3       | 55.7 | 63.1     | 20.0 | 28.4  | 26.4 | 33.9 |
|                | StandardRAG [14] | Llama3/E5    | 58.9 | 68.3     | 25.1 | 35.3  | 8.6  | 21.0 |
|                | SEAKR [48]       | Llama2/BM25  | 42.6 | 48.6     | 27.9 | 39.7  | 30.2 | 36.0 |
|                | DRAGIN [35]      | Llama2/BM25  | 54.4 | 62.3     | 23.7 | 34.2  | 22.4 | 30.0 |
|                | Learned          |              |      |          |      |       |      |      |
|                | Self-RAG [1]*    | Llama3/E5    | 38.2 | 53.4     | 17.1 | 29.6  | 12.1 | 25.1 |
| Auto-RAG [49]* | Llama3/E5        | 56.7         | 60.7 | 31.9     | 44.3 | 38.8  | 46.9 |      |
| (b)            | Prompting        |              |      |          |      |       |      |      |
|                | NaiveGen [14]    | GPT-4        | 59.6 | 70.0     | 27.5 | 37.9  | 23.7 | 31.7 |
|                | StandardRAG [14] | GPT-4/BM25   | 60.3 | 65.9     | 35.7 | 45.5  | 28.7 | 33.8 |
|                | FLARE [13]*      | Llama3/E5    | 55.8 | 63.2     | 19.7 | 28.0  | 25.8 | 33.9 |
|                | IR-CoT [38]*     | Llama3/BM25  | 56.4 | 68.9     | 28.6 | 39.1  | 23.5 | 31.8 |
|                | Learned          |              |      |          |      |       |      |      |
|                | SIM-RAG (ours)   | Llama-3/BM25 | 70.7 | 75.6     | 32.7 | 43.3  | 34.1 | 40.2 |
| SIM-RAG (ours) | GPT-4/BM25       | 77.5         | 82.7 | 39.8     | 52.2 | 46.1  | 54.6 |      |

# When Less Is More: Standard RAG Beats Complex Methods—SIM-RAG is the Exception

|                  |                  | TriviaQA     |           |
|------------------|------------------|--------------|-----------|
| Method           | LLM/Retriever    | EM           | F1        |
| <i>Prompting</i> |                  |              |           |
| (a)              | NaiveGen [14]    | Llama3       | 55.7 63.1 |
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|                  | SIM-RAG (ours)   | Llama-3/BM25 | 70.7 75.6 |
|                  | SIM-RAG (ours)   | GPT-4/BM25   | 77.5 82.7 |

# Learned Small Critic Model Beats GPT-4 Critic on Complex Tasks

SIM-RAG (GPT-4) with using different Critic models



# Knowing When to Say No: Example of SIM

**Question:** The character Jack from While You Were Sleeping was born in what year?

**LLM:** 1964. According to the film...

**Critic rejects answer.**

**Query:** birth year of Sandra Bullock

**Retrieved Document:** Sandra Annette Bullock (born July 26, 1964) is an American actress...

**LLM:** 1964. The question is asking about Sandra...

**Critic rejects answer.**

**Query:** birth year of Sandra Bullock's co-star in While You Were Sleeping

**Retrieved Document:** While You Were Sleeping (film) stars Sandra Bullock as Lucy, a Chicago Transit Authority token collector, and Bill Pullman as Jack the brother of a man...

**LLM:** 1950. Since Jack is Sandra Bullock's co-star, we can infer that Jack is likely an adult in the film, given the age difference between Sandra Bullock and Bill Pullman. Therefore, it is likely that Jack was born in the 1950s.

**Critic rejects answer.**

**Query:** birth year of Bill Pullman

**Retrieved Document:** William James "Bill" Pullman (born December 17, 1953) is an American...

**LLM:** 1953

**Critic accepts answer.**

# Summary: Self-Practice Inner Monologue (SIM) Matter

SIM-RAG:

- Efficient and effective add on for any RAG system

Looking ahead:

- Inner Monologue Learning: SIM enables self-practice beyond RAG — a cost-effective way to generate fresh training data with intermediate reasoning steps.

**Open Source:** [github.com/ucscirkm/SIM-RAG](https://github.com/ucscirkm/SIM-RAG)

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Thank You!  
Q&A



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